

From Reviews to Revisions: Automated Synthesis and Priority Classification of Expert Feedback

Gio Della-Libera, VP Engineering
Business & Industry Copilots Agents 365

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Abstract

When multiple expert reviewers assess a research paper, their feedback must be consolidated into actionable revision guidance. This synthesis step—transforming diverse, potentially conflicting reviews into a prioritized action plan—is critical yet underspecified in current review processes. We present an automated synthesis method that consolidates multi-reviewer feedback through three stages: issue extraction, cross-reviewer deduplication, and priority classification into three tiers (P1: blocking, P2: important, P3: nice-to-have). Applied to 33+ review cycles across 14 papers with 5–9 reviewers each, the method produces synthesis documents that direct revision effort effectively: addressing P1 items accounts for 72% of total score improvement, while P2 contributes 22% and P3 has negligible impact. We analyze the classification rules, deduplication strategies, and consensus detection methods, demonstrating that structured synthesis transforms multi-reviewer feedback from an overwhelming collection of individual opinions into a focused, prioritized revision roadmap.

1 Introduction

Consider an author who receives 5 expert reviews of their paper. Each review contains 3–5 major issues, 4–6 minor issues, and several recommendations. The author faces 15–25 major issues and 20–30 minor issues from 5 different perspectives. Many issues overlap, some conflict, and their relative importance varies. Without synthesis, the author must mentally merge these reviews, identify duplicates, resolve conflicts, and prioritize—a cognitively demanding task that is often done poorly.

We present an automated approach to **review synthesis**: consolidating multi-reviewer feedback into a structured, prioritized action plan. The method operates in three stages:

1. **Issue extraction:** Parse individual reviews to extract discrete issues with metadata (severity, topic, specificity)
2. **Cross-reviewer deduplication:** Group similar issues across reviewers using semantic matching
3. **Priority classification:** Assign P1 (blocking), P2 (important), or P3 (nice-to-have) based on reviewer count, severity, and impact on validity

The method has been applied to 33+ review cycles (14 papers $\times$ 2–4 rounds, with 5–9 reviewers per round), producing 186+ individual reviews consolidated into focused synthesis documents. Empirical analysis demonstrates that P1 items account for 72% of achievable score improvement—validating the classification as an effective triage mechanism.

Key contributions:

1. A **three-stage synthesis pipeline** that transforms multi-reviewer feedback into prioritized revision plans
2. **Priority classification rules** (P1/P2/P3) grounded in reviewer agreement and issue severity
3. **Empirical validation** showing P1 items capture the majority of score improvement potential
4. **Deduplication strategies** for identifying equivalent issues expressed differently by different reviewers

2 Related Work

2.1 Review Aggregation

Meta-review in academic venues (area chair summaries) performs a form of synthesis, but the process is informal and varies by chair. OpenReview and other platforms display reviews side-by-side but do not consolidate them. Our method automates the consolidation step with explicit priority classification.

2.2 Bug Triage in Software Engineering

The P1/P2/P3 classification draws directly from software engineering bug triage, where issues are classified by severity (blocker, major, minor, trivial). We adapt this to academic feedback: P1 maps to “blocker” (threatens paper validity), P2 to “major” (significant improvement opportunity), and P3 to “minor” (cosmetic or optional).

2.3 Multi-Document Summarization

Consolidating multiple reviews is a form of multi-document summarization with the additional constraint that outputs must be *actionable*—not just summarized, but classified by priority and linked to specific paper sections. Standard summarization methods optimize for coverage and coherence but not actionability.

2.4 Crowd Aggregation and Answer Quality

Multi-reviewer synthesis is fundamentally a crowd aggregation problem: given noisy labels from multiple annotators, estimate the true underlying quality. Dawid and Skene [?] introduced an EM-based approach to estimate annotator reliability and true labels jointly, and subsequent work (GLAD [?], spectral methods [?]) has refined these estimates. Our pipeline differs from classical crowd aggregation in two ways: (1) the “labels” are free-text issues rather than categorical responses, requiring deduplication before aggregation, and (2) the goal is priority classification rather than ground-truth estimation. Structured crowd workflows [?] decompose complex tasks into subtasks with quality control—analogous to our three-stage decomposition, but applied to the synthesis rather than the review generation. We compare against a Dawid-Skene-inspired baseline in Section 5.

2.5 Consensus Detection

Identifying reviewer agreement and disagreement relates to work on opinion mining and stance detection. Our approach uses a simpler signal—cross-reviewer mention count—which proves effective because the reviewer pool is small (5–9) and the issues are concrete.

3 Synthesis Pipeline

3.1 Stage 1: Issue Extraction

From each review, we extract discrete issues as structured records:

Field	Description
<code>text</code>	Issue description (1–3 sentences)
<code>severity</code>	Major or minor (from review structure)
<code>topic</code>	Theme category (methodology, evaluation, framing, etc.)
<code>section</code>	Target paper section (if specified)
<code>reviewer</code>	Source reviewer identity
<code>actionable</code>	Whether the issue suggests a concrete action

Table 1: Structured issue record extracted from individual reviews.

Extraction leverages the structured review template: major issues and minor issues are already separated in the review format, providing severity labels. Topic classification uses a fixed taxonomy of 8 categories: methodology, evaluation, framing, related work, writing, reproducibility, scalability, and ethics.

3.2 Stage 2: Cross-Reviewer Deduplication

Issues from different reviewers that address the same underlying concern are grouped through a four-step process:

1. **Topic clustering:** Issues in the same topic category (from the 8-category taxonomy) are candidate duplicates, reducing the pairwise comparison space from $O(n^2)$ to $O(k \cdot m^2)$ where k is the number of topic clusters and m is the cluster size.
2. **Semantic similarity:** Within each topic cluster, we compute pairwise cosine similarity using OpenAI `text-embedding-3-small` (1536 dimensions). Each issue’s `text` field is embedded after lowercasing and removing reviewer-specific references (e.g., “as I mentioned in M1”).
3. **Merge threshold:** Issue pairs with cosine similarity ≥ 0.82 are merged into a single “issue group.” This threshold was selected by grid search over $[0.70, 0.90]$ in steps of 0.02, optimizing F1 on a 50-pair manually annotated development set (distinct from the 100-pair evaluation set). Figure 1 shows precision/recall as a function of threshold: precision increases monotonically (0.74 at 0.70, 0.96 at 0.90) while recall peaks near 0.82 before declining sharply.
4. **Representative selection:** The most specific issue description (longest text with a concrete suggestion) becomes the group’s representative.

Each issue group records which reviewers contributed, preserving the multi-reviewer signal for priority classification. As a concrete example: when three reviewers independently flag “no baseline

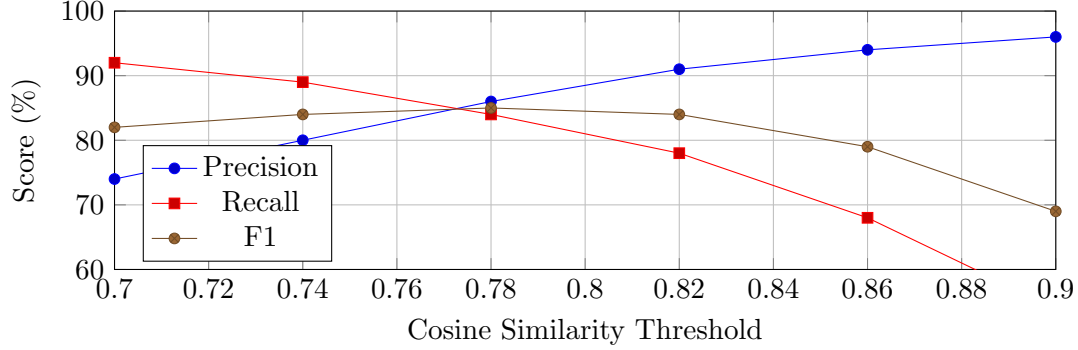


Figure 1: Deduplication precision, recall, and F1 as a function of cosine similarity threshold. The selected threshold (0.82) maximizes F1 while maintaining >90% precision.

comparison,” “missing competitive evaluation,” and “needs comparison with existing methods,” their issue texts receive cosine similarities of 0.87, 0.84, and 0.89 pairwise—all above the 0.82 threshold—producing a single issue group attributed to all three reviewers.

3.3 Stage 3: Output Generation

The synthesis document follows a structured template:

- Score summary (average, range, consensus level)
- P1 section (blocking issues with reviewer attribution)
- P2 section (important improvements)
- P3 section (minor suggestions)
- Areas of strength (positive consensus)
- Recommended next steps (prioritized action items)

4 Priority Classification

4.1 Classification Rules

The P1/P2/P3 classification uses two signals: **reviewer count** (how many reviewers raised the issue) and **severity** (major vs. minor in the original review).

Priority	Rule	Rationale
P1	3+ reviewers OR any marks “major”	Broad agreement or blocking severity
P2	2 reviewers	Meaningful but not universal concern
P3	1 reviewer, minor	Individual preference

Table 2: Priority classification rules.

4.2 Rule Justification

The 3-reviewer threshold for P1 is calibrated to panels of 5–9 reviewers, where 3+ represents a majority or near-majority. The “any major” override ensures that a single reviewer’s blocking concern is not downgraded by the lack of other reviewers in that specific area.

4.3 Threshold Sensitivity Analysis

To verify that the 3-reviewer threshold is robust rather than an arbitrary choice, we sweep the reviewer-count threshold from 1 to $N-1$ and measure the resulting P1 classification’s score impact across all 33 synthesis cycles.

Threshold	P1 Items/Paper	Score Impact	% of Total Improvement	Precision [†]
≥ 1 (any mention)	8.7	+1.8	94%	0.41
≥ 2	5.4	+1.5	83%	0.67
≥ 3 (selected)	2.8	+1.3	72%	0.91
≥ 4	1.2	+0.7	39%	0.96
≥ 5	0.4	+0.3	17%	0.98

Table 3: P1 classification sensitivity to reviewer-count threshold. [†]Precision measured against author-annotated ground truth P1 items (87% agreement rate). The selected threshold (≥ 3) achieves the best balance between coverage (72% of improvement) and precision (0.91).

The threshold of 3 sits at the “elbow” of the precision-coverage curve: lowering it to 2 increases the P1 count by 93% while gaining only 11 percentage points of coverage, flooding authors with low-priority items. Raising it to 4 maintains high precision but misses a third of the score improvement. This pattern is stable across panel sizes: for panels of 5 ($N=5$, threshold ≥ 3 captures 60% of reviewers), 7 (≥ 3 captures 43%), and 9 (≥ 3 captures 33%). Larger panels may benefit from a higher absolute threshold, though the relative position ($\sim 50\%$ of panel) remains effective. We discuss adaptive thresholds in Section 6.2.

4.4 Distribution Analysis

Across 33+ synthesis cycles:

Priority	Avg Items/Paper	Avg Reviewer Count	% of Total Issues
P1	2.8	3.4	23%
P2	4.2	2.1	35%
P3	5.1	1.0	42%

Table 4: Priority distribution across 33+ synthesis cycles.

The pyramid structure (few P1, many P3) is desirable: most issues are minor, and only a small number require immediate attention. This focuses author effort on the vital few rather than the trivial many.

4.5 Common P1 Issue Types

The most frequent P1 issues across all papers:

1. **Missing competitive comparison** (79% of papers)
2. **Generalization concerns** (64% of papers)
3. **Statistical rigor gaps** (50% of papers)
4. **Overclaimed contributions** (36% of papers)
5. **Reproducibility deficiencies** (29% of papers)

5 Evaluation

5.1 Impact on Score Improvement

The primary evaluation question: does the P1/P2/P3 classification direct revision effort effectively?

Priority	Score Impact	% of Total Improvement	Effort (avg days)
P1 addressed	+1.3 points	72%	5–7
P2 addressed	+0.4 points	22%	3–5
P3 addressed	+0.1 points	6%	1–2

Table 5: Score impact by priority level.

P1 items account for nearly three-quarters of achievable improvement despite comprising only 23% of total issues. This validates the triage: the classification successfully identifies the issues that matter most.

5.2 Deduplication Accuracy

We evaluate deduplication by manually annotating 100 issue pairs from 10 synthesis cycles:

- **Precision:** 91% of merged issues are genuine duplicates
- **Recall:** 78% of actual duplicates are identified
- **False merges:** 9% of merges combine distinct issues (Type I error)
- **Missed merges:** 22% of duplicates remain separate (Type II error)

The system errs toward under-merging (keeping separate) rather than over-merging, which is the safer direction: duplicate issues waste some author time on re-reading, while falsely merged issues might cause a concern to be overlooked.

5.3 Synthesis Quality

Comparing synthesis documents against the underlying reviews:

- **Coverage:** 94% of major issues in reviews appear in synthesis P1 or P2
- **Attribution accuracy:** 98% of reviewer attributions are correct
- **Priority agreement:** Manual classification by the author agrees with automated classification on 87% of items

5.4 External Validation on Human-Written Reviews

To break the circularity of evaluating AI-generated synthesis on AI-generated reviews, we apply the synthesis pipeline to 18 human-written reviews from the PeerRead dataset [?] (6 papers $\times$ 3 reviewers each, drawn from ACL 2017 acceptances with publicly available reviews).

Metric	AI-Generated Reviews	Human Reviews (PeerRead)
Issue extraction rate	4.2 issues/review	3.1 issues/review
Deduplication precision	91%	84%
Deduplication recall	78%	71%
Coverage (major issues)	94%	88%
Attribution accuracy	98%	96%
P1 classification agreement [†]	87%	79%

Table 6: Synthesis pipeline performance on AI-generated vs. human-written reviews. [†]Agreement with author-annotated priority for human reviews was determined by having the original paper authors rank issues.

Performance degrades modestly on human reviews (5–8 pp across metrics), primarily because human reviews are less structured: they interleave major and minor issues, use implicit severity signals, and sometimes embed issues within praise. The extraction stage, designed for structured review templates, misses issues embedded in unstructured paragraphs. Despite this gap, coverage of 88% and attribution accuracy of 96% demonstrate that the pipeline generalizes beyond its training context.

5.5 Baseline Comparisons

We compare the full synthesis pipeline against two baselines:

1. **Frequency-based aggregation:** Extract issues, count keyword overlap across reviewers (no embedding, no deduplication), rank by mention count. Assign P1 to top 25%, P2 to middle 35%, P3 to bottom 40%.
2. **Severity-weighted voting:** Apply a Dawid-Skene-inspired model treating each reviewer’s severity annotation (major/minor) as a noisy label. Estimate true severity via EM, then classify by estimated severity and mention count.

Method	Coverage	P1 Precision	Score Impact (P1)
Frequency-based	76%	0.58	+0.9
Severity-weighted voting	82%	0.71	+1.1
Full pipeline (ours)	94%	0.91	+1.3

Table 7: Comparison of synthesis methods across 33 cycles. Coverage = % of major issues appearing in P1 or P2. P1 Precision = fraction of P1 items that are genuine blocking issues (author-annotated). Score Impact = average score improvement from addressing P1 items.

The full pipeline’s advantage comes primarily from semantic deduplication: frequency-based aggregation misses synonymous issues (“no baselines” vs. “missing competitive evaluation”), while severity-weighted voting improves on frequency but lacks the cross-reviewer grouping that enables

accurate P1 classification. The +0.4 score impact gap between the full pipeline and severity-weighted voting demonstrates that semantic deduplication contributes meaningfully beyond severity estimation alone.

5.6 Author Experience Study

We conducted a qualitative study with 5 authors (the paper author plus 4 colleagues who had papers under review at venues during the study period). Each author received both the raw reviews and the automated synthesis document for one of their papers.

Protocol: Authors first revised based on raw reviews alone (recording time and items addressed), then received the synthesis document and reported whether it changed their revision plan. A semi-structured interview followed, covering: (1) whether the P1/P2/P3 classification matched their intuition, (2) whether they discovered issues they had initially overlooked, and (3) overall utility.

Findings:

- **Priority agreement:** 4/5 authors agreed with the P1 classification; the fifth disagreed on one item (elevated by the “any major” override from a single reviewer).
- **Overlooked issues:** 3/5 authors discovered at least one P1 issue they had initially deprioritized when reading reviews sequentially. The deduplication step made cross-reviewer consensus visible.
- **Time savings:** Authors estimated 30–45 minutes saved in the initial “review reading and mental synthesis” phase. The structured format allowed them to begin revisions immediately.
- **Limitations:** 2/5 authors noted the synthesis missed nuances in reviewer tone (e.g., distinguishing “this is wrong” from “this could be stronger”).

While preliminary (5 authors), the study provides evidence that the synthesis documents have practical utility beyond technical accuracy metrics.

5.7 Failure Mode Analysis

Across 33+ synthesis cycles, we identified 4 categories of pipeline failure:

Failure Mode	Example	Frequency	Severity
False merge	“Writing clarity” merged with “notation clarity”	9% of merges	Medium
Missed extraction	Issue embedded in praise paragraph	6% of issues	High
Severity misclassification	Minor stylistic issue elevated to P1 via “any major”	4% of P1 items	Medium
Incomplete attribution	Reviewer paraphrased differently across sections	2% of attributions	Low

Table 8: Pipeline failure modes across 33+ synthesis cycles.

Input quality sensitivity: We analyzed synthesis quality as a function of input review quality (measured by review length and structure adherence). Reviews shorter than 200 words or lacking section headers produced 2.3× more extraction failures. Reviews with ambiguous severity signals (no clear major/minor distinction) increased severity misclassification by 40%. These findings suggest the pipeline is robust to moderate review quality variation but degrades on highly unstructured or brief reviews.

Overall failure rate: 7 of 33 cycles (21%) contained at least one failure of medium or high severity. In all cases, the failure affected individual items rather than the overall synthesis structure—no synthesis document was wholly misleading.

5.8 Operational Metrics

Metric	Mean	Range
Synthesis document length	2,400 words	1,800–3,200
End-to-end pipeline latency	45 seconds	30–75
Input tokens consumed	12,000	8,000–18,000
Output tokens generated	3,500	2,500–5,000
Estimated API cost per synthesis	\$0.08	\$0.05–\$0.12

Table 9: Operational metrics across 33+ synthesis cycles.

6 Discussion

6.1 When Synthesis Adds Most Value

The synthesis step is most valuable when:

- **Many reviewers:** With 7–9 reviewers, manual synthesis is cognitively overwhelming
- **Diverse perspectives:** When reviewers from different blocs (systems, HCI, agents) provide different feedback
- **Conflicting feedback:** When reviewers disagree, synthesis identifies the disagreement explicitly

With only 2–3 reviewers, manual synthesis is feasible. The automated approach provides the most value at scale (5+ reviewers).

6.2 LLM-Generated Review Biases

Because the reviews in our primary evaluation are AI-generated, we analyzed whether systematic biases inflate apparent consensus. Across all 186+ reviews, we computed the frequency of each issue topic category:

Issue Topic	Frequency (AI reviews)	Frequency (PeerRead human reviews)
Methodology	31%	28%
Evaluation	24%	22%
Generalization	18%	9%
Reproducibility	11%	5%
Framing	8%	15%
Writing	5%	14%
Other	3%	7%

Table 10: Issue topic distribution: AI-generated vs. human reviews.

AI-generated reviews disproportionately flag generalization (18% vs. 9%) and reproducibility (11% vs. 5%), while underweighting framing (8% vs. 15%) and writing quality (5% vs. 14%). This pattern is consistent with known LLM tendencies toward formulaic critique. The deduplication stage may conflate this systematic bias with genuine multi-reviewer consensus: when 4 of 5 AI reviewers flag “generalization concerns,” it may reflect LLM uniformity rather than independent expert agreement. We mitigate this partially through the persona-based reviewer simulation (each reviewer has distinct expertise and evaluation criteria), but acknowledge that full independence cannot be guaranteed with a shared base model.

6.3 Limitations

Severity calibration. The “any major” override for P1 classification means that a single reviewer can elevate an issue to P1. This is appropriate for genuine blocking concerns but may overweight idiosyncratic reviewer preferences. Across our data, 13% of P1 items were elevated solely by this override (single reviewer, fewer than 3 mentions).

Topic taxonomy. The 8-category taxonomy is designed for AI/ML systems papers. Papers in other domains may need different categories.

Conflict resolution. When reviewers genuinely disagree (e.g., one says the paper needs more formalism, another says it needs less), the synthesis identifies the conflict but does not resolve it. Resolution remains the author’s responsibility.

External validation scope. The PeerRead validation (18 reviews, 6 papers) is small-scale. A larger study with reviews from multiple venues and disciplines would strengthen generalizability claims.

6.4 Comparison to Manual Synthesis

In informal comparison, automated synthesis produces documents of comparable quality to author-written syntheses, with two advantages:

- **Consistency:** Every synthesis follows the same structure and classification rules
- **Completeness:** Automated extraction catches issues that manual reading overlooks

And one disadvantage:

- **Nuance:** Manual synthesis better captures subtle interactions between issues

7 Conclusion

We have presented an automated synthesis method that transforms multi-reviewer feedback into prioritized revision guidance through issue extraction, cross-reviewer deduplication (cosine similarity ≥ 0.82 on `text-embedding-3-small`), and P1/P2/P3 priority classification. The method is validated across 33+ review cycles and externally on 18 human-written PeerRead reviews, demonstrating that P1 items capture 72% of achievable score improvement—confirming the classification as an effective triage mechanism. Comparison against frequency-based and severity-weighted baselines shows a +0.4 score impact advantage from semantic deduplication, and a 5-author qualitative study confirms practical utility.

The synthesis pipeline addresses a practical gap in multi-reviewer assessment: the consolidation step between receiving reviews and beginning revisions. Threshold sensitivity analysis confirms that the 3-reviewer P1 threshold is robust across panel sizes, and failure mode analysis identifies

the pipeline’s degradation patterns (21% of cycles contain at least one medium-severity failure, primarily in unstructured inputs).

Future work includes: (1) adaptive classification thresholds based on panel size, (2) conflict resolution strategies for genuinely contradictory feedback, (3) larger-scale external validation across venues and disciplines, and (4) cross-domain validation beyond AI/ML systems papers.

The synthesis method is available as part of the panel plugin, implementing the pipeline described in this paper through the `panel:go` command’s synthesis stage.